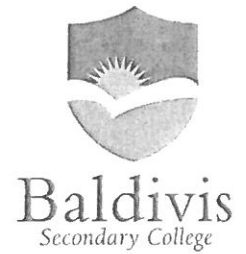


Year 10
Pathway 2



Mini Test 5 solving linear and quadratic equations

Name: _____

Score: /26
Total Time: 40 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Scientific calculators may be used and 1 A4 page of notes will be allowed.

Solve the following problems

1. $x + 4 = 7$ $x = 3$ ✓ [1]

2. $t - 5 = 16$ $t = 21$ ✓ [1]

3. $5m = 35$ $m = 7$ ✓ [1]

4. $\frac{y}{4} = 7$ $y = 28$ ✓ [1]

5. $5x + 3 = 23$ $x = 4$ ✓ [1]

6. $6k - 4 = -40$ $k = -6$ ✓ [1]

7. $\frac{x+1}{2} = 9$ $x = 17$ ✓ [1]

8. $-2(f + 5) = 10$ [1]

$-2f - 10 = 10$
 $-2f = 20$
 $f = -10$ ✓

9. $(x+5)(x+2) = 0$ $x = -5$ ✓ or $x = -2$ ✓ [2]

10. $x(x+2) = 0$ $x = 0$ ✓ or $x = -2$ ✓ [2]

11. $4x(x-7) = 0$ $x = 0$ ✓ or $x = 7$ ✓ [2]

12. $(3x+5)(x+2) = 0$ $3x+5=0$ ✓ $x+2=0$
 $x = -\frac{5}{3}$ or $x = -2$ ✓ [2]

13. $x(2-x) = 0$ $x = 0$ ✓ or $x = 2$ ✓ [2]

14. $x^2 = 36$ $x = \pm 6$ ✓ ✓ [2]

15. $x^2 + 6x + 8 = 0$ [3]

$(x+4)(x+2) = 0$ ✓

$x = -4$ or $x = -2$ ✓ ✓

16. $x^2 - 10x + 25 = 0$ [3]

$(x-5)(x-5) = 0$ ✓

$x = 5$ ✓ ✓